



ELEVVO INTERNSHIP 2025 (TASK-05)

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Exploratory Data Analysis on Titanic Dataset

Introduction

The Titanic dataset is a classic dataset widely used in data analysis and machine learning. The main objective of this task was to perform Exploratory Data Analysis (EDA) to understand survival patterns based on different passenger attributes.

Objectives

- Perform data cleaning by handling missing values and encoding categorical features.
- Generate summary statistics for key numerical features.
- Explore group-based survival insights such as by gender, passenger class, and age.
- Visualize survival patterns using plots and correlation heatmaps.
- Save the cleaned dataset and group outputs for future analysis.

Steps Performed

1. Data Cleaning

- Filled missing Age values with the median.
- Filled missing Embarked values with the most frequent value.
- Dropped the Cabin column due to a large number of missing values.
- Encoded categorical variables (Sex and Embarked) into numeric form.
- Checked for duplicates and confirmed there were none.

- Saved the cleaned dataset as “Titanic_Cleaned.csv”.
- 2. **Feature Engineering**
 - Created FamilySize = SibSp + Parch + 1.
 - Created IsAlone flag to identify passengers traveling alone.
 - Created AgeBand to categorize passengers into Child, Teen, YoungAdult, Adult, and Senior.
- 3. **Summary Statistics**
 - Generated descriptive statistics for Age, Fare, SibSp, Parch, and FamilySize.
- 4. **Group-Based Insights**
 - Survival by Sex: Females had a survival rate of around 74% compared to only 19% for males.
 - Survival by Passenger Class: First-class passengers had the highest survival rate, while third-class had the lowest.
 - Survival by Age Band: Children had the highest survival rate, while older adults had lower chances of survival.
- 5. **Visualizations**
 - Bar plots of survival count, survival by sex, passenger class, and embarked port.
 - Histograms of Age and Fare distributions by survival status.
 - Correlation heatmap to identify relationships between numerical features.
- 6. **Exported Outputs**
 - Saved group insights as CSV files: survival_by_sex.csv, survival_by_pclass.csv, and survival_by_ageband.csv.

Deliverables

- Cleaned dataset: Titanic_Cleaned.csv
- Group insights CSV files
- Visualizations showing survival trends and correlation heatmap
- Final report summarizing findings

Key Findings

- Gender played a major role: women were much more likely to survive than men.
- Passenger class strongly influenced survival, with first-class passengers having the highest chances.
- Children had better survival chances compared to teens and adults.
- Fare and passenger class showed a strong correlation with survival outcomes.

Conclusion

This task successfully cleaned the Titanic dataset, created new meaningful features, and provided valuable insights through group-based analysis and visualizations. The analysis clearly shows that gender, passenger class, and age were important factors that influenced survival during the Titanic disaster.

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